

LATENCY-AWARE SELECTIVE MULTI-VIEW RETRIEVAL FOR UNSUPERVISED SOCIAL EVENT CLUSTERING

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ABSTRACT

Streaming social event monitoring requires timely event assignments. Existing retrieval-augmented detectors rely on text queries or invoke multiple evidence views unconditionally, causing query drift and long remote-LLM critical paths. We propose MvSED, which treats multi-view evidence as a bounded residual correction. Text retrieval fixes candidate membership, relevance–redundancy selection chooses rerankers, and a frozen label-free gate accepts only consistent repairs of uncertain assignments. On Event2012, MvSED yields mean NMI, AMI, and ARI changes of +0.00023, +0.00005, and +0.00069, while reducing runtime and LLM calls by 11.3% and 1.3%. On Event2018, the corresponding mean changes are +0.0098, +0.0127, and +0.0567, with runtime and calls reduced by 22.3% and 5.9%. Although Event2018 tokens increase by 11.9%, failed attempts fall by 53.7%, exposing a practical latency–token trade-off.

Index Terms— social event monitoring, short-text clustering, retrieval-augmented generation, multi-view retrieval, end-to-end latency

1. INTRODUCTION

Social event detection groups a stream of short social-media messages into evolving real-world events. It supports crisis monitoring and public-opinion analysis, where an accurate cluster delivered late may already describe a stale situation [1, 2]. Moreover, event memory is updated sequentially, so delayed assignments postpone the context available to later blocks. Timeliness is therefore part of the monitoring objective, not merely an implementation statistic. The task remains difficult because messages are sparse, informal, and lexically diverse; local graph structure alone may not associate different reports of the same event [3, 4].

Retrieval-augmented generation (RAG) supplies complementary context from an external memory [5]. RagSEDE applies this principle to unsupervised social event detection and event-memory evolution [6]. A single text query can miss an informative entity, relation, or temporal pattern, whereas applying every view to every anchor causes query drift and

extra inference. For remotely served LLMs, token volume is not a sufficient proxy for elapsed time: serial calls, failures and retries, retrieval, and service delay all lie on the critical path. Spending more tokens in a better-grounded interaction can therefore be reasonable if it avoids slower repeated interactions.

We address this trade-off with MvSED, which targets lower wall-clock latency subject to preserving clustering quality, rather than minimum token count. Multi-view retrieval is a *selective residual correction*: the text view retrieves the only admissible candidate set; only uncertain assignments activate entity, relation, and temporal reranking; and a deterministic gate accepts a repair only under semantic-support and ranking-margin constraints. Confident decisions bypass auxiliary reasoning, while ambiguous anchors spend additional evidence tokens on a bounded repair path.

Our core contribution is a latency-aware selective residual-retrieval design. Text fixes candidate membership, relevance–redundancy selection admits only useful auxiliary rerankers, and a frozen label-free gate changes only uncertain assignments supported by consistent evidence. Under identical anchor units, this single design reduces paired runtime by 11.3–22.3%; Event2012 mean metric changes remain below 0.001, while Event2018 shows favorable observed mean changes.

2. RELATED WORK

Social event detection. Early methods model lexical, user, or topic signals [7, 8]; later systems use graphs, incremental learning, and structural entropy [3, 9, 4]. PromptSED uses label-assisted prompt learning and LSED a label-supervised classifier [10, 11], so their published scores are not mixed with our frozen label-free-inference comparison. We include HISEvent [4] as an external unsupervised quality reference. RagSEDE is the controlled baseline because MvSED retains its anchors, detector, LLM, and memory protocol, changing only residual retrieval and repair.

Retrieval and multi-view reasoning. Dense retrieval and RAG provide external evidence for downstream reason-

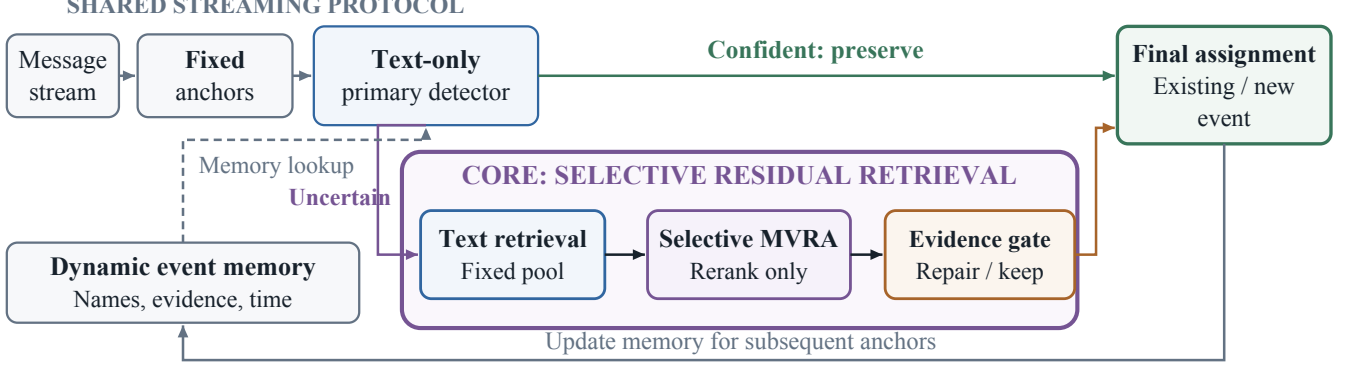


Fig. 1. Focused overview of MvSED. Shared stream processing and memory are peripheral; the emphasized core admits only uncertain primary outputs to text-grounded selective reranking and frozen evidence-gated repair.

ing [12, 5]. Query rewriting can improve recall but may drift from the original intent [13]. Multi-view learning likewise requires explicit control of noisy and redundant views [14, 15]. MvSED combines these observations by separating candidate generation, auxiliary reranking, and repair acceptance.

3. METHOD

Let $\{m_j\}_{j=1}^N$ denote the messages in block t and $\mathcal{M}_{t-1}$ the event memory accumulated from previous blocks. The task assigns an open-set event identifier to every message. For policy π , the deployment objective is

$$\min_{\pi} T_{e2e}(\pi) \quad \text{s.t.} \quad Q(\pi) \geq Q_{\text{base}} - \epsilon, \quad (1)$$

where T_{e2e} is the time until a block is committed to memory and Q is clustering quality. We report token volume separately rather than treating it as a monotonic latency surrogate. Figure 1 summarizes the complete path. Inference never accesses task labels, although the pretrained LLM contributes learned prior knowledge.

3.1. Protocol-aligned anchors and memory

We adopt RagSEDE’s single-stage anchor protocol to keep the inference units identical across methods. Sentence-BERT embeddings [16] are grouped using a fixed cosine threshold τ_a . We denote the anchor capacity by B_a , the representative-message budget by R , and the representativeness–redundancy trade-off by ρ . These quantities are selected during development and then frozen. No second-stage anchor reclustering is applied. Figure 2 illustrates the chronological construction and representative selection.

For each event, $\mathcal{M}_{t-1}$ stores its name, keywords, linked anchors, representative messages, and time observations. A text-only primary detector maps anchor a_i to an existing event

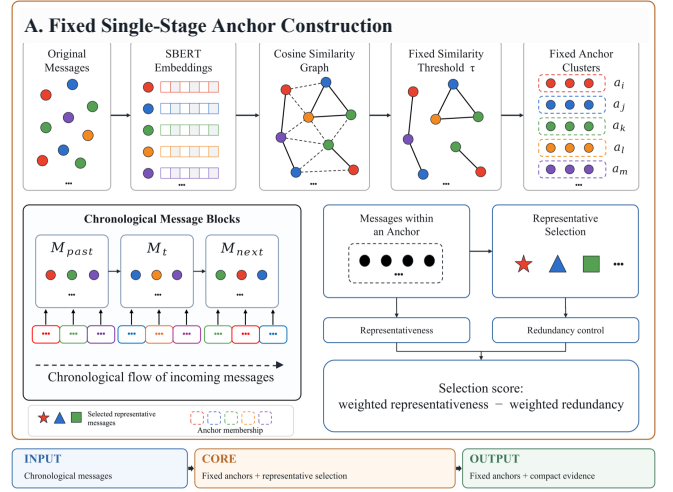


Fig. 2. Shared fixed-anchor protocol used by both methods. Chronological messages are embedded and grouped once with a fixed threshold; representative messages retain coverage while controlling redundancy.

$e \in \mathcal{M}_{t-1}$ or to OTHERS. The final decision is used to update memory.

3.2. Stage-aware selective multi-view retrieval

The selected representatives of a_i are concatenated and normalized to form the text query $q_i^{(t)}$. Figure 3 shows how this query retrieves the only admissible candidate set

$$\mathcal{C}_i = \text{TopK}_{K_c} \left(r_t(q_i^{(t)}, e), \mathcal{M}_{t-1} \right). \quad (2)$$

where r_t is raw SBERT cosine similarity. Auxiliary views $v \in \{\text{entity, relation, time}\}$ are generated from the same anchor and cannot introduce events outside $\mathcal{C}_i$. The fixed-pool

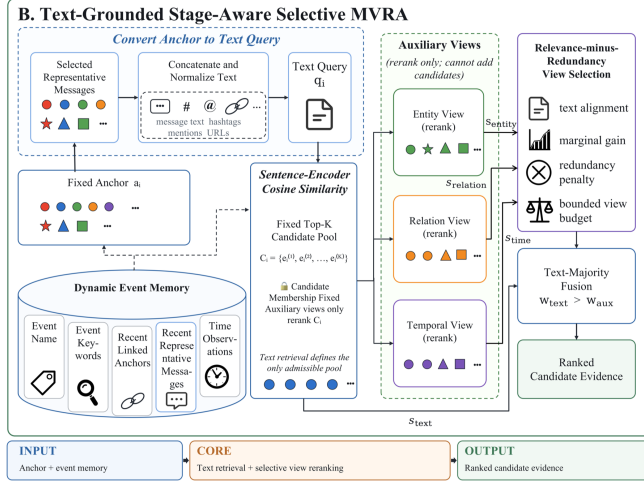


Fig. 3. Text-grounded selective MVRA. Text retrieval fixes candidate membership; entity, relation, and temporal views may only rerank that pool before text-majority fusion.

constraint limits query drift to a reranking operation.

Let $\tilde{s}_v$ be view v 's normalized score vector over $\mathcal{C}_i$. Text is always retained, and at most $L - 1$ auxiliary views are selected. For auxiliary view v , define its centered ranking-profile alignment A_v to text, redundancy R_v , score dispersion σ_v , and top-candidate consistency c_v by

$$A_v = \text{corr}(\tilde{s}_v, \tilde{s}_t), \quad R_v = \max_{u \in S_a} |\text{corr}(\tilde{s}_v, \tilde{s}_u)|, \quad (3)$$

$$g(v) = \sigma_v [c_v \max(0, A_v) - \lambda R_v].$$

A view is retained only if $A_v \geq \theta_{\text{align}}$ and $g(v) \geq \theta_{\text{gain}}$. Entity and relation views provide structural support. Time may resolve a ranking tie but cannot independently justify a repair. Candidate scores are fused with a text majority,

$$s_f(e) = \omega_t \tilde{s}_t(e) + (1 - \omega_t) \sum_{v \in S_a} \alpha_v \tilde{s}_v(e), \quad (4)$$

where $\alpha_v = g(v) / \sum_{u \in S_a} g(u)$; if S_a is empty, $s_f = \tilde{s}_t$. The asymmetric weight $\omega_t > 1 - \omega_t$ preserves text as primary evidence.

3.3. Uncertainty-gated residual repair

The multi-view path is invoked only when the primary detector returns *OTHERS* or *Confusion*. Figure 4 summarizes this residual path. Let e^* and e° denote the highest-scoring candidate and its runner-up, respectively. The gate accepts a reassignment to e^* if

$$G_i = [n_{\text{view}} \geq \eta_{\text{view}}] \wedge [n_{\text{struct}} \geq \eta_{\text{struct}}] \\ \wedge [r_t(q_i^{(t)}, e^*) \geq \theta_{\text{text}}] \wedge [s_f(e^*) \geq \theta_{\text{fuse}}] \\ \wedge [s_f(e^*) - s_f(e^\circ) \geq \theta_{\text{margin}}]. \quad (5)$$

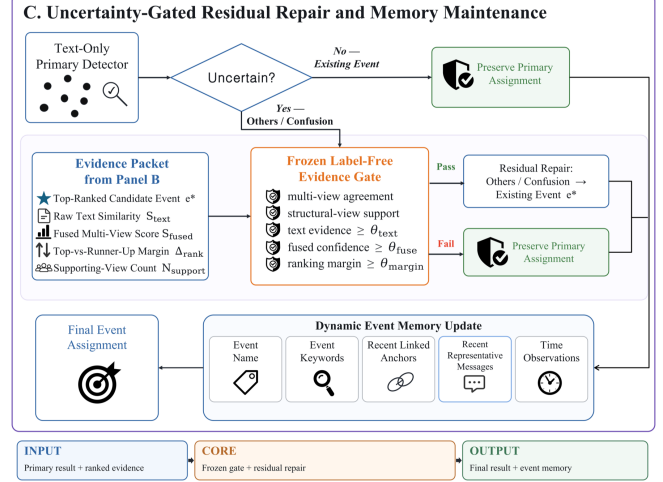


Fig. 4. Uncertainty-gated residual repair. Confident primary decisions bypass repair; uncertain decisions are changed only when the frozen evidence gate passes, after which event memory is updated.

View v supports e when e lies in its top- K_s ranking over $\mathcal{C}_i$. Here, n_{view} counts gate-eligible text, entity, and relation support, whereas n_{struct} counts entity and relation support only; time is excluded from both counts. A failed gate restores the primary output, whereas a passed gate deterministically assigns e^* ; the reported method issues no second decision-stage LLM call. Gate parameters are selected under the development protocol and frozen before testing. Held-out labels never enter inference and are consulted only post hoc for evaluation.

4. EXPERIMENTS

4.1. Experimental setup

Data and split. Event2012 contains 68,841 English tweets from 503 events collected over four weeks [17]; Event2018 contains 64,516 French tweets from 257 events over 23 days [18]. Following the incremental protocol, we evaluate 21 Event2012 blocks and 16 Event2018 blocks. Event2012 M1–M6 are used for method development. View-selection and gate parameters are then frozen for Event2012 M7–M21 and used unchanged on Event2018; only the pre-specified dataset encoder and fixed anchor threshold differ.

Models. Event2012 and Event2018 use all-MiniLM-L6-v2 and multilingual DistilUSE encoders, respectively. Both methods query the same DeepSeek-R1-Distill-Qwen-32B RAGFlow endpoint on one RTX 4090 (24 GB). **Metrics.** We report NMI, AMI, and ARI [19]; paired end-to-end time is the primary efficiency endpoint, with calls, failures, LLM time, tokens, and peak memory as diagnostics. **Protocol.** Both methods share blocks, encoders, LLM, prompts, memory

Table 1. Quality comparison. HISEvent[†] averages RagSEDE’s published two-decimal scores [6] over matching test blocks and is not seed/system matched.

Method	Event2012			Event2018		
	NMI	AMI	ARI	NMI	AMI	ARI
HISEvent [†]	0.80200	0.79200	0.61133	0.75375	0.74688	0.60375
RagSEDE	0.89637	0.88610	0.78386	0.75418	0.74012	0.61094
MvSED	0.89660	0.88615	0.78455	0.76400	0.75284	0.66762

Table 2. Paired efficiency (both datasets) and aligned Event2018 ablation (R/M: RagSEDE/MvSED; bold: best value in each ablation column).

Data	End-to-end latency			LLM calls		Failed attempts	
	R/M (h)	Reduction	Speed-up	R/M	Reduction	R/M	Reduction
Event2012	11.15/9.89	11.3%	1.13×	2702/2668	1.3%	128/71	44.5%
Event2018	23.22/18.04	22.3%	1.29×	4749/4468	5.9%	123/57	53.7%
Combined	34.37/27.93	18.7%	1.23×	7451/7136	4.2%	251/128	49.0%
<i>Aligned Event2018 ablation (M1–M16)</i>							
Method	NMI	AMI	ARI	Proc. h	Calls	Failures	Time red.
RagSEDE	0.75418	0.74012	0.61094	23.12	4749	123	—
MVRA	0.77229	0.76044	0.65022	20.46	4753	80	11.49%
NoTime	0.75317	0.74126	0.63830	17.06	4467	63	26.18%
MvSED	0.76400	0.75284	0.66762	17.93	4468	57	22.45%

state, and ordered anchors and representatives. We admit a paired run only if its anchor-audit hashes match; equal counts alone do not suffice.

Frozen implementation. Anchor thresholds are 0.40 and 0.30 for Event2012 and Event2018. Each anchor contains at most 100 messages and three representatives. We set $K_c = 8$, $L = 3$, text/auxiliary weights to 0.65/0.35, alignment/gain thresholds to 0.05/0.015, and $\lambda = 0.35$. The gate requires two views, one structural view, text/fused scores of 0.45/0.40, and a 0.03 margin. Generation permits 2048 tokens, 30 attempts, and a 15-s retry delay. All values are fixed during testing; code and audit artifacts will be released.

4.2. Controlled quality and efficiency

Table 1 reports the controlled RagSEDE comparison and an external HISEvent reference. HISEvent is averaged from RagSEDE’s published two-decimal block scores over the same test blocks, rather than rerun under our seed and system protocol. On held-out Event2012, the three-seed MvSED–RagSEDE changes are +0.00023, +0.00005, and +0.00069 for NMI, AMI, and ARI. Event2018 has favorable fixed-run changes of 0.0098, 0.0127, and 0.0567, winning 10, 11, and 11 of 16 blocks. All block-bootstrap 95% intervals include zero.

The upper panel of Table 2 shows that MvSED is faster on 14/15 Event2012 and 16/16 Event2018 blocks; bootstrap latency-reduction intervals are 6.9–15.8% and 18.3–26.6%. Calls fall by 1.3% and 5.9%, and failures by 44.5% and 53.7%. On Event2018, 5.18 h less end-to-end time coincides with 5.26 h less LLM time and 66 fewer failures. Retries add provider waiting, backoff, and reissuing to the serial path; retrieval and embedding partly offset the saving. This is aggregate attribution, not block-level causality.

4.3. Aligned ablation

On the same 16 fixed-anchor Event2018 blocks, MVRA alone reaches the highest mean NMI/AMI (0.77229/0.76044), improves ARI by 0.03928 over RagSEDE, and reduces aggregate processing time by 11.49%. NoTime is the fastest variant: mean paired processing time falls by 25.47% (95% CI: 15.10–33.97%; 14/16 wins; sign $p = .0042$), with a 26.18% aggregate reduction. The full model attains the highest ARI (0.66762) and fewest failed attempts (57) while remaining 22.45% faster in aggregate; its paired mean reduction is 21.65% (CI: 17.26–25.93%; 16/16 wins; $p = .000031$). Restoring temporal evidence over NoTime yields favorable observed NMI/AMI/ARI changes of +0.01083/+0.01158/+0.02932 at 5.06% aggregate overhead. All reported quality intervals cross zero, so these quality effects are directional rather than statistically significant.

5. CONCLUSION

We presented MvSED, whose core contribution is a text-grounded selective residual path: a fixed candidate pool bounds auxiliary reranking, and a frozen gate changes only uncertain assignments. Event2012 mean metric changes are below 0.001; Event2018 has favorable observed fixed-run changes of 0.0098, 0.0127, and 0.0567. Runtime falls by 11.3% on Event2012 and 22.3% on Event2018 (18.7% combined, 1.23×). On Event2018, 11.9% more tokens coexist with fewer calls and failures and 24.5% less LLM time, so latency and token volume must be reported separately.

We evaluate two Twitter benchmarks; remote load makes latency deployment-specific. Quality intervals include zero, so we claim neither significance nor equivalence. Labels enter development and post-hoc evaluation, never test inference. Because Event2018 tokens increase, we claim observed latency reduction, not lower cost, energy, or universal efficiency; other platforms remain untested.

AI-use disclosure. Codex assisted language editing; all text was author-verified.

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